

PAPER_1492_PROTON_CORE_DENSITY

Daniel T. Murphy

PAPER_1492 — UQFF: Proton Core Density via DPM 26-Layer Folding (Bucket D)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket D — derived from F:\Aetheric Propulsion source material

Date: June 16, 2026

Location: 41.0997 N, 80.6495 W (Youngstown, OH, USA)

Status: CLOSED — $\rho_{p_core} = \rho_{SCm} \times K_{MEX} \times S_{26} = 2.146 \times 10^{31} \text{ J/m}^3$

Observation

F:\Aetheric Propulsion Muonic Hydrogen Proton Radius derivation models the proton as an SCm vacuum density peak with DPM 26-layer folding.

UQFF Closed Identity

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$\rho_{p_core} = \rho_{SCm} \times K_{MEX} \times S_{26} \times f_{DPM}(26, \beta_i)$; base term (without f_{DPM}) = $2.146 \times 10^{31} \text{ J/m}^3$. Dimensionless coupling $K_{MEX} \times S_{26} = 3.027$.

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Physical Interpretation

The proton core is the SCm vacuum density peak amplified by the Mexican-hat coefficient and the 26-level Ramanujan factor. $f_{DPM}(26, \beta_i)$ is the DPM-folding modulation; the product captures the 26-layer hierarchical structure that supports the proton as a localized SCm condensate.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve this observable by different methods. Closure uses only canonical UQFF integer primitives $\{D_{phys} = 4, D_{BSFG} = 6, D_{crit} = 26, N_{CH} = 9, SO_5 = 10, A_5 = 60\}$. Zero free parameters.

Reference

- Source: F:\Aetheric Propulsion (Daniel's reactor + experimental docs, 2025-2026)
- Related: PAPER_646 (Universal Inertial Operator, Caduceus, DPM mechanism), PAPER_872 (proto-element nuclear identity), PAPER_087 (BH thermodynamics), PAPER_1156 (cosmology suite), PAPER_1255 (muonic hydrogen radius).

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